

Practice Sheet: Random Variable

1. A random variable X has the following probability function:

x	-2	-1	0	1	2	3
$p(x)$	0.1	k	0.2	$2k$	0.3	k

- (i) Find the value of k .
 - (ii) Construct the c.d.f. $F(x)$ and draw its graph.
2. A random variable X has the following probability distribution:

x	0	1	2	3	4	5	6	7	8
$p(x)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

- (i) Determine the value of a .
 - (ii) Find $P(X < 3)$, $P(X \geq 3)$, and $P(0 < X < 5)$.
 - (iii) What is the smallest value of x for which $P(X \leq x) > 0.5$?
 - (iv) Find the distribution function of X .
3. Let $p(x)$ be the probability function of a discrete random variable X which assumes the values x_1, x_2, x_3, x_4 such that $2p(x_1) = 3p(x_2) = p(x_3) = 5p(x_4)$. Find the probability distribution and the cumulative distribution function X .
4. The following is the distribution function of a discrete random variable X :

x	-1	0	1	2	3	5	8
$F_X(x)$	0.10	0.30	0.45	0.50	0.75	0.90	1.00

- (i) Find the probability distribution of X .
 - (ii) Find $P(X \text{ is even})$ and $P(1 \leq X \leq 8)$.
 - (iii) Find $P(X = -3 \mid X < 0)$ and $P(X \leq 3 \mid X > 0)$.
5. Suppose that the random variable X assumes three values 0, 1, and 2 with probabilities $\frac{1}{6}$, $\frac{1}{2}$, and $\frac{1}{3}$ respectively. Obtain the distribution function of X .
6. A continuous random variable X follows the probability law: $f(x) = Ax^2, 0 \leq x \leq 1$. Determine A and find the probability that
- (i) X lies between 0.2 and 0.5,
 - (ii) X is less than 0.3,
 - (iii) $\frac{1}{4} < X < \frac{1}{2}$ and
 - (iv) $X > \frac{3}{4}$ given $X > \frac{1}{2}$.
7. For the probability density function: $f(x) = cx^2(1-x), 0 < x < 1$, find the constant c .

8. A petrol pump is supplied with petrol once a day. If its daily volume of sales (X) in thousands of liters is distributed by: $f(x) = 5(1 - x)^4, 0 \leq x \leq 1$. What must be the capacity of its tank in order that the probability that its supply will be exhausted in a given day shall be 0.01?
9. The length (in hours) X of a certain type of light bulb may be supposed to be a continuous random variable with *p.d.f.*:
- $$f(x) = \begin{cases} \frac{a}{x^3}, & 1500 < x < 2500 \\ 0, & \text{elsewhere.} \end{cases}$$
- Determine the constant a , the distribution function of X , and compute the probability of the event $1700 \leq X \leq 1900$.
10. A continuous random variable X has the distribution function:
- $$F(x) = \begin{cases} 0, & x \leq 1 \\ k(x - 1)^4, & 1 < x \leq 3 \\ 1, & x > 3 \end{cases}$$
- Find k , and the probability density function $f(x)$.
11. A random variable X has the probability mass function:
- | | | | | | |
|--------|-----|------|------|------|-----|
| x | 0 | 1 | 2 | 3 | 4 |
| $p(x)$ | k | $2k$ | $3k$ | $2k$ | k |
- (i) Find k . (ii) Compute $P(X \geq 2)$. (iii) Find the cumulative distribution function (c.d.f.) and sketch its graph.
12. The distribution function of a discrete random variable X is
- $$F(x) = \begin{cases} 0, & x < 0 \\ 0.2, & 0 \leq x < 1 \\ 0.5, & 1 \leq x < 2 \\ 0.8, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$
- (i) Determine the probability mass function of X .
(ii) Find $P(X = 2)$, $P(X > 1)$.
13. A continuous random variable X has probability density function $f(x) = kx^3, 0 \leq x \leq 2$.
(i) Find k .
(ii) Compute $P(0.5 \leq X \leq 1.5)$.
(iii) Find the c.d.f. of X .
14. Let the lifetime (in hours) of a device be a continuous random variable with density $f(x) = \frac{1}{100}e^{-x/100}, x > 0$.
(i) Find the distribution function $F(x)$.
(ii) Compute $P(X > 200)$.
(iii) Find the median lifetime.
15. A continuous random variable X has cumulative distribution function
- $$F(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-\lambda x}, & x > 0 \end{cases}$$

- (i) Find the corresponding p.d.f.
 - (ii) Show that it is a valid density.
 - (iii) Compute $P(1 \leq X \leq 3)$.
16. The daily demand (in units) for a product is a discrete random variable with pmf $p(x) = \frac{6-x}{15}$, $x = 1, 2, 3, 4, 5$.
- (i) Find $P(X \leq 2)$.
 - (ii) Find the expected demand.
 - (iii) Construct the c.d.f.
17. Suppose a continuous random variable X has pdf $f(x) = \frac{2}{x^3}$, $x > 1$.
- (i) Verify that this is a valid pdf.
 - (ii) Find $P(2 \leq X \leq 4)$.
 - (iii) Obtain the c.d.f.
18. The random variable X represents the number of heads in three tosses of a fair coin.
- (i) Write down the probability distribution of X .
 - (ii) Find $P(X \geq 2)$.
 - (iii) Construct and graph the c.d.f. of X .